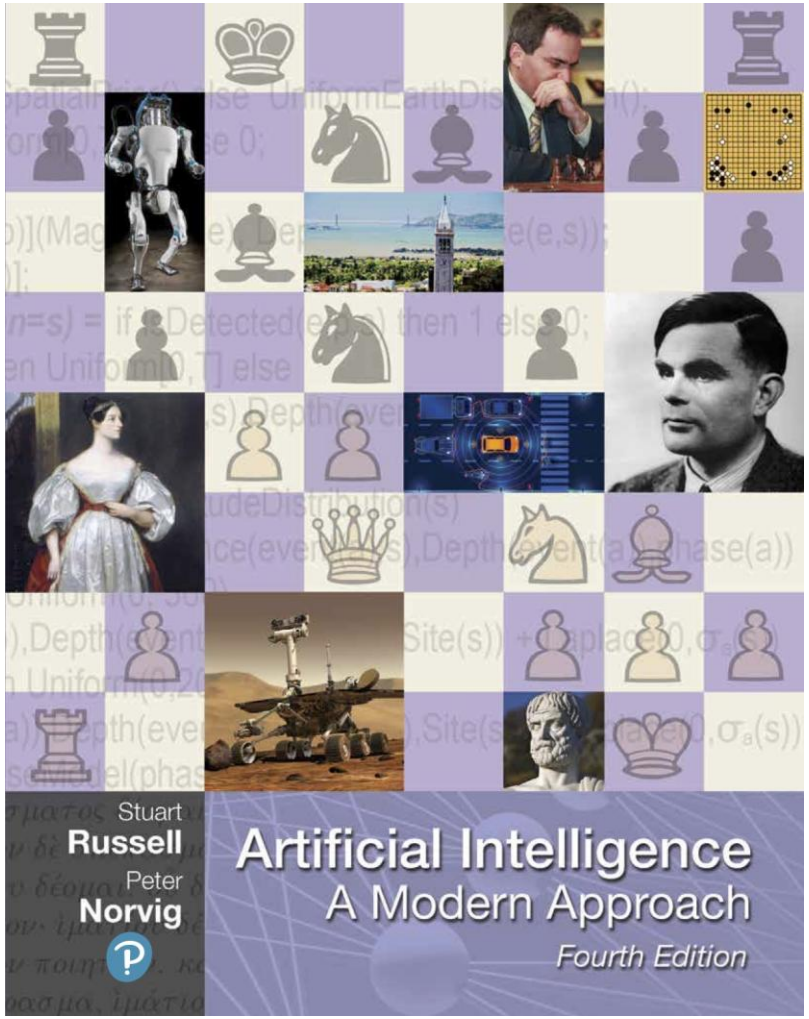
AI Foundations for Engineers

AIFE-FDP

Unit – 1 (07 Hours)

Introduction: What is AI? Acting humanly: The Turing test approach,
Thinking humanly: The cognitive modelling approach,
Thinking rationally: The “laws of thought” approach,
Acting rationally: The rational agent approach;
The foundations of AI: Mathematics, Economics, Neuroscience, Psychology, Computer Engineering; The State of the Art; Risks and Benefits of AI

Intelligent Agents: Agents and Environments, The concept of Rationality, The Nature of Environments



Artificial Intelligence?

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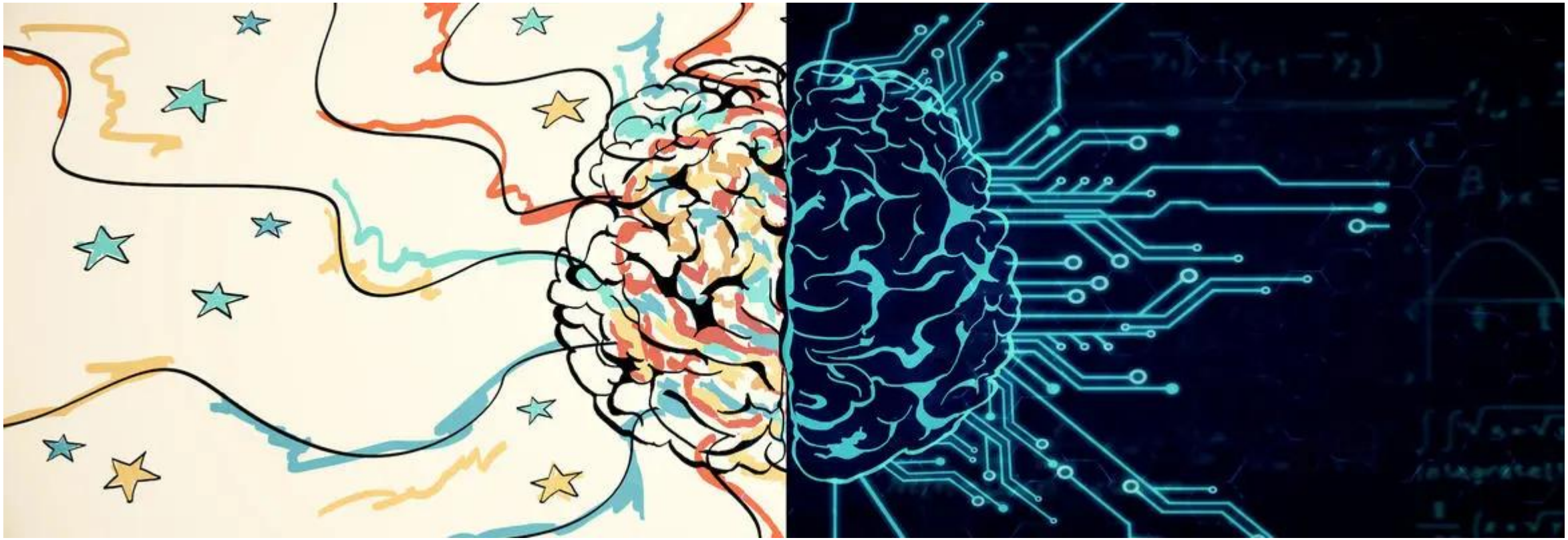

Artificial Intelligence

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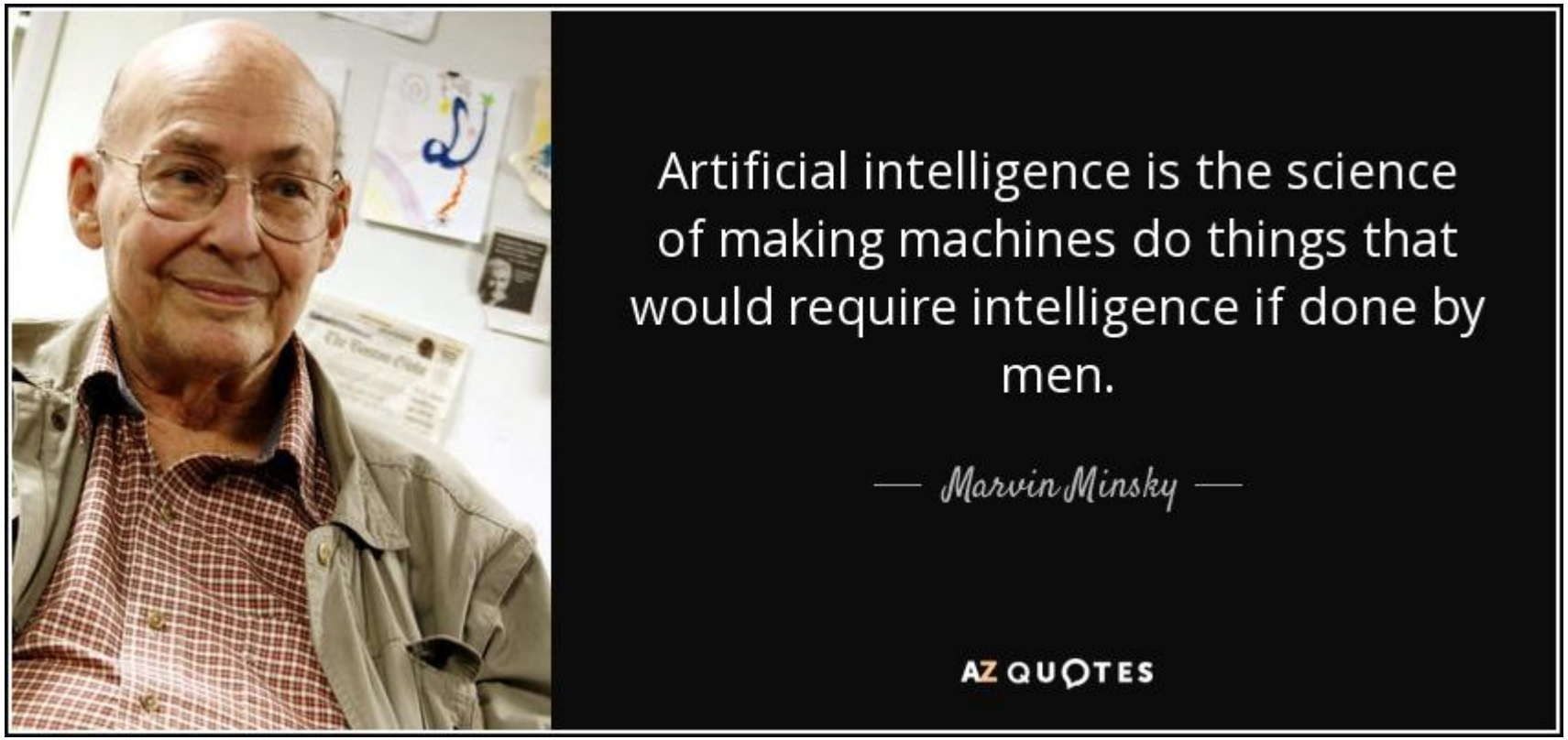

Marvin Minsky (1927-2016) was a pioneering cognitive scientist and co-founder of the MIT Media Lab, widely regarded as one of the founding fathers of artificial intelligence.

AI Focus

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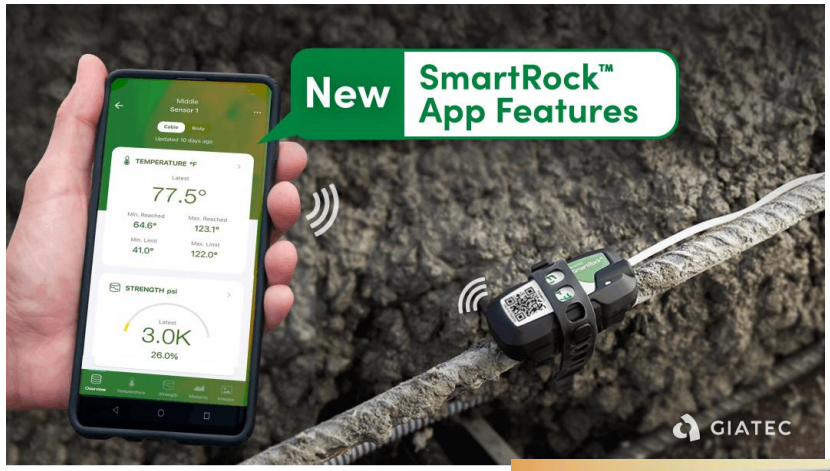
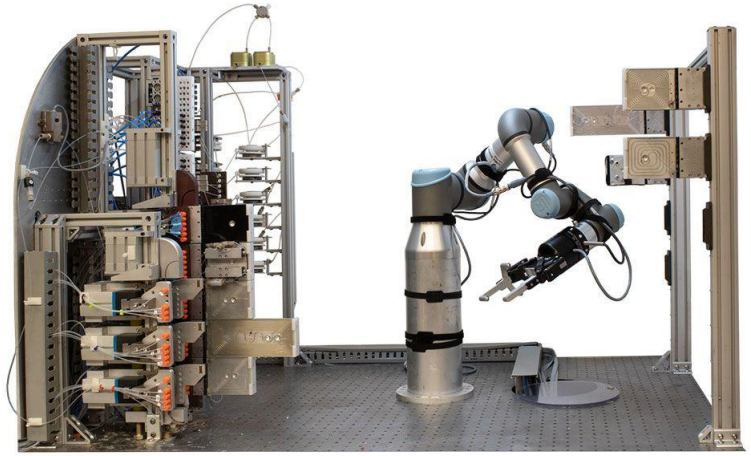
- AI focuses on building intelligent machines that can effectively and safely navigate **novel situations.**





Acting humanly: The Turing test approach

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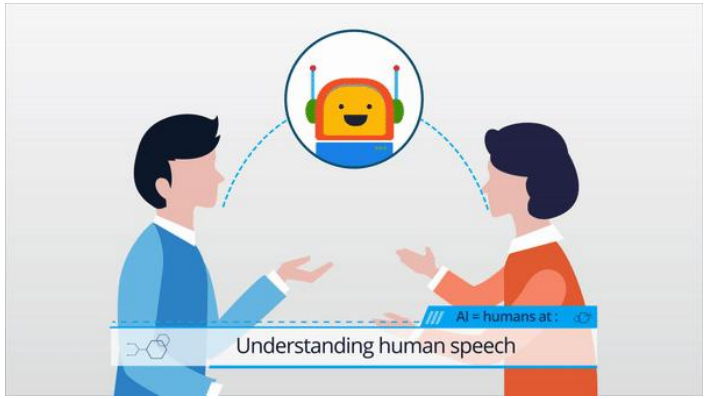
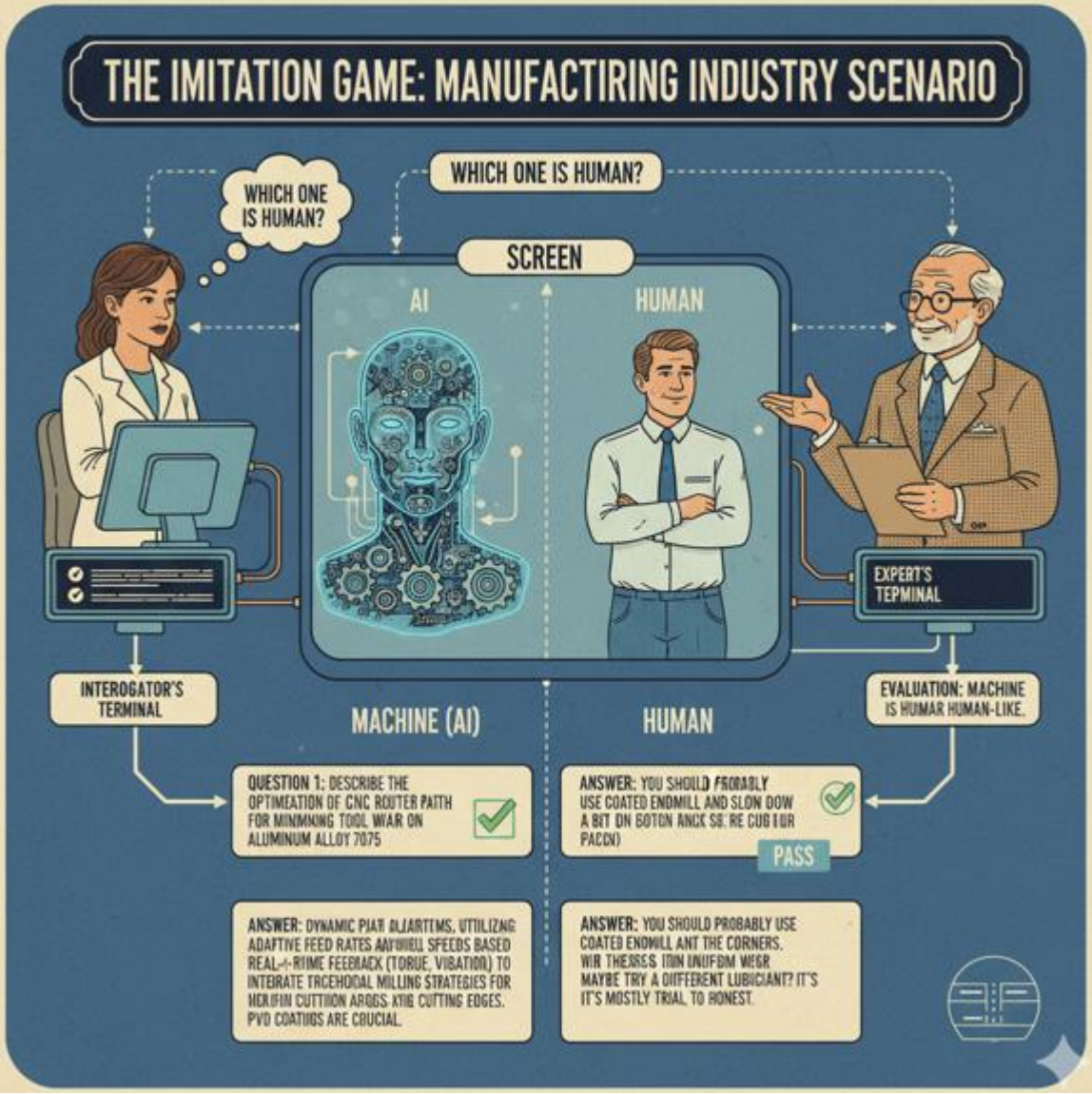
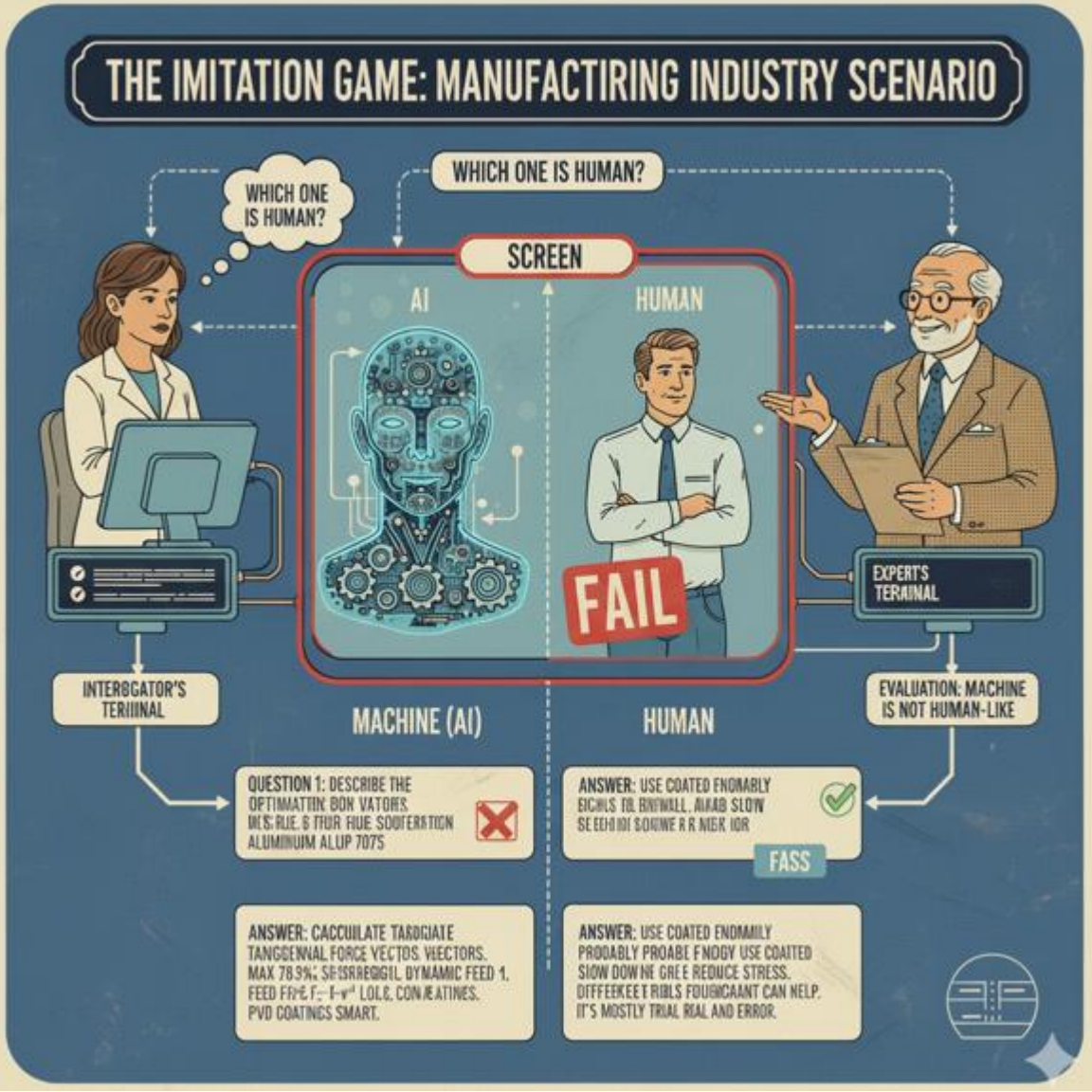
- The Turing test, proposed by Alan Turing (1950), was a thought experiment => **"Can a machine think?"**
- The computer would need the following capabilities:
  - Natural language processing for human communication
  - Knowledge representation for information storage
  - Automated reasoning for answering questions and drawing conclusions
  - Machine learning for adaptation and pattern recognition




THE IMITATION GAME: MANUFACTIRING INDUSTRY SCENARIO
WHICH ONE IS HUMAN?
WHICH ONE IS HUMAN?
SCREEN
AI
HUMAN
EXPERT'S TEPMINAL
INTEROGATOR'S TERMINAL
MACHINE (AI)
HUMAN
EVALUATION: MACHINE IS HUMAR HUMAN-LIKE.
QUESTION 1: DESCRIBE THE OPTIMEATION OF CNC ROUTER PATH FOR MINMNING TOOL WIAR ON ALUMINUM ALLOY 7075
ANSWER: YOU SHOULD PROBABLY USE COATED ENDMILL AND SLOW DOW
PASS
ANSWER: DYNAMIC PIAT ALJARTTMS, UTTILIZNG ADAFTIVE FEED RATES AAFUIIELL SPEEDS BASED REAL-TIIME FEEBRACK (TORUE, VIBATION) TO INTEBRATE TRCEHODAL MILLING STRATEGIES FOR HERIFIN CUTTIION ARDGS XTIG CUTTING EDGES. PVD COATIIGS ARE CBUCIAL.
ANSWER: YOU SHOULD PROBABLY USE COATED ENDMILL ANT THE CORNERS. WIR THESSS TOIN UNUFBM WEGR MAYBE TRY A DIFFERENT LUBICIANT? IT'S IT'S MOSTLY TRIAL TO HONEST.


THE IMITATION GAME: MANUFACTIRING INDUSTRY SCENARIO
WHICH ONE IS HUMAN?
WHICH ONE IS HUMAN?
SCREEN
AI
HUMAN
FAIL
EXPERT'S TERAINAL
INTERBGATOR'S TERIIINAL
MACHINE (AI)
HUMAN
EVALUATION: MACHINE IS NOT HUMAN-LIKE
QUESTION 1: DESCRIBE THE OPTIMMATTIN BON VATOFES ALUMIINUIM ALUP 7075
ANSWER: USE COATED ENOMALLY
FASS
ANSWER: CACCUILATE TARDGIATE TANGGENNAL FORCE VECTOS VIECTORS. MAX 78.9%; SEISSREGGIL DYNAMIC FEED 1, PVD COATINGS SMART.
ANSWER: USE COATED ENOMALLY PROBABLY PROABE FNOGY USE COATED SIOW DOWNE GREE REDUCE STRESS. DIFFEEKEET RIBLS FOUIICAANT CAN NELP. IT'S MOSTLY TRIAL REAL AND ERROR.

Thinking humanly: The cognitive modeling approach

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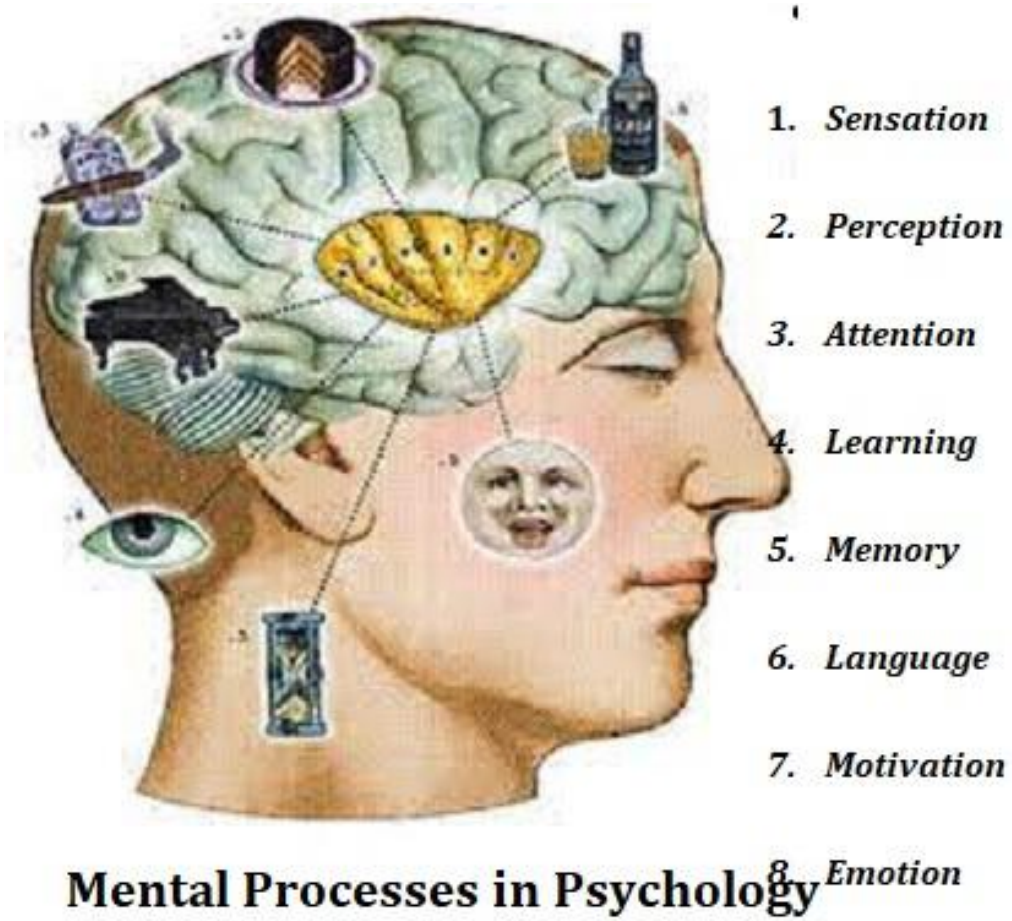

Thinking humanly : The cognitive modeling approach

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- **Introspection**—trying to catch our thoughts as they go by;

- **Feeling Anxious**:

"My heart's racing… I'm thinking the worst… Is the situation really dangerous, or is my mind exaggerating?"

Thinking humanly: The cognitive modeling approach

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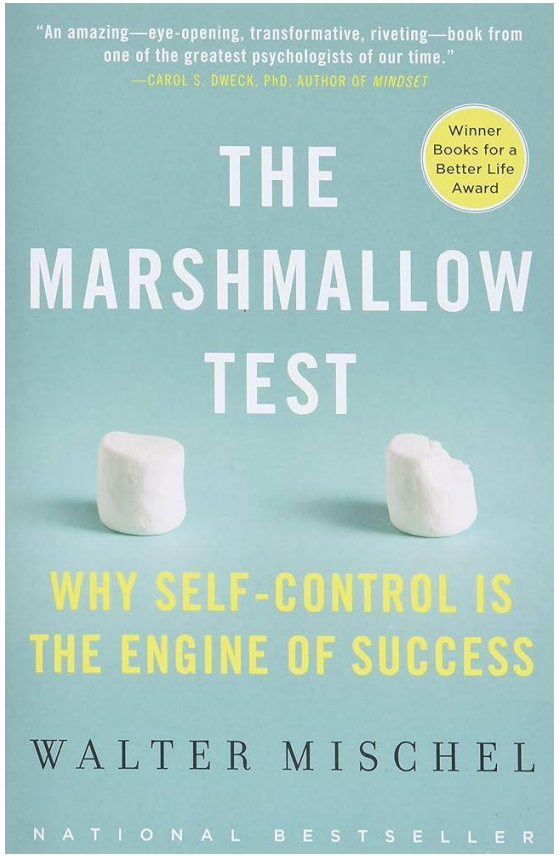
- **Psychological experiments**—observing a person in action;



Thinking humanly: The cognitive modeling approach

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- **Brain imaging** —observing the brain in action.

Thinking humanly: The cognitive modeling approach

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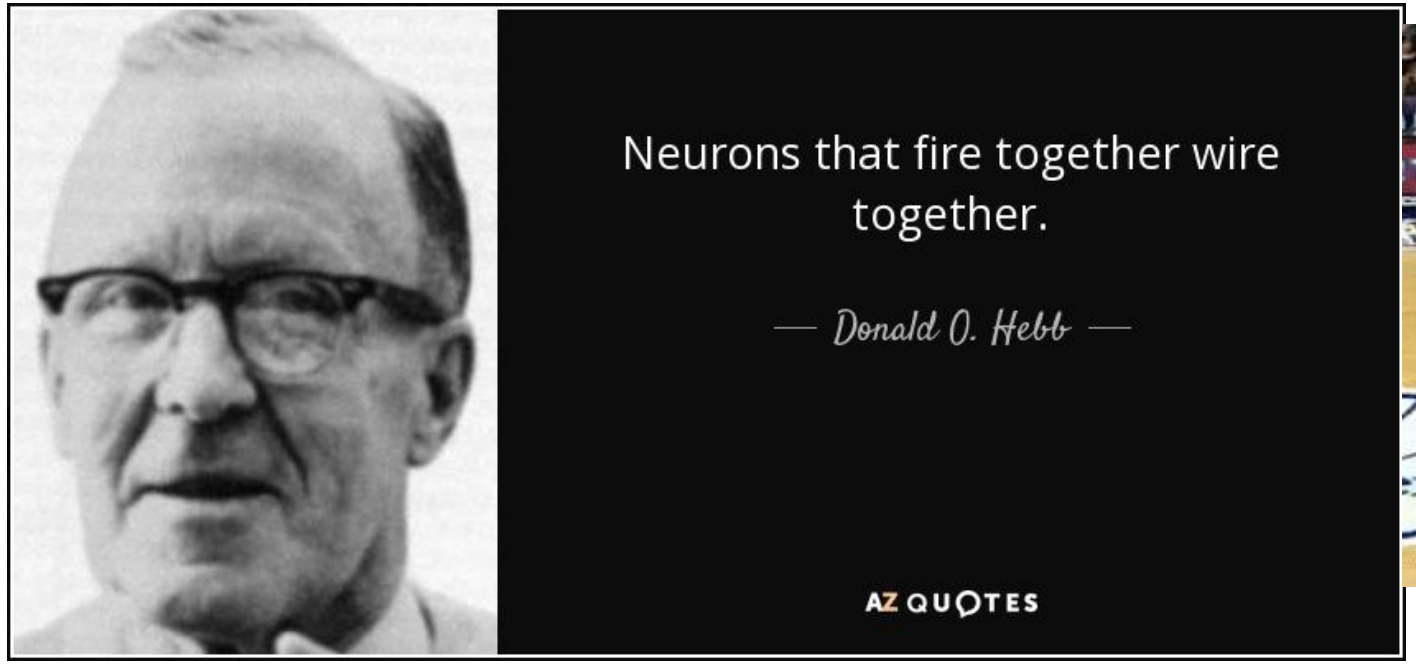
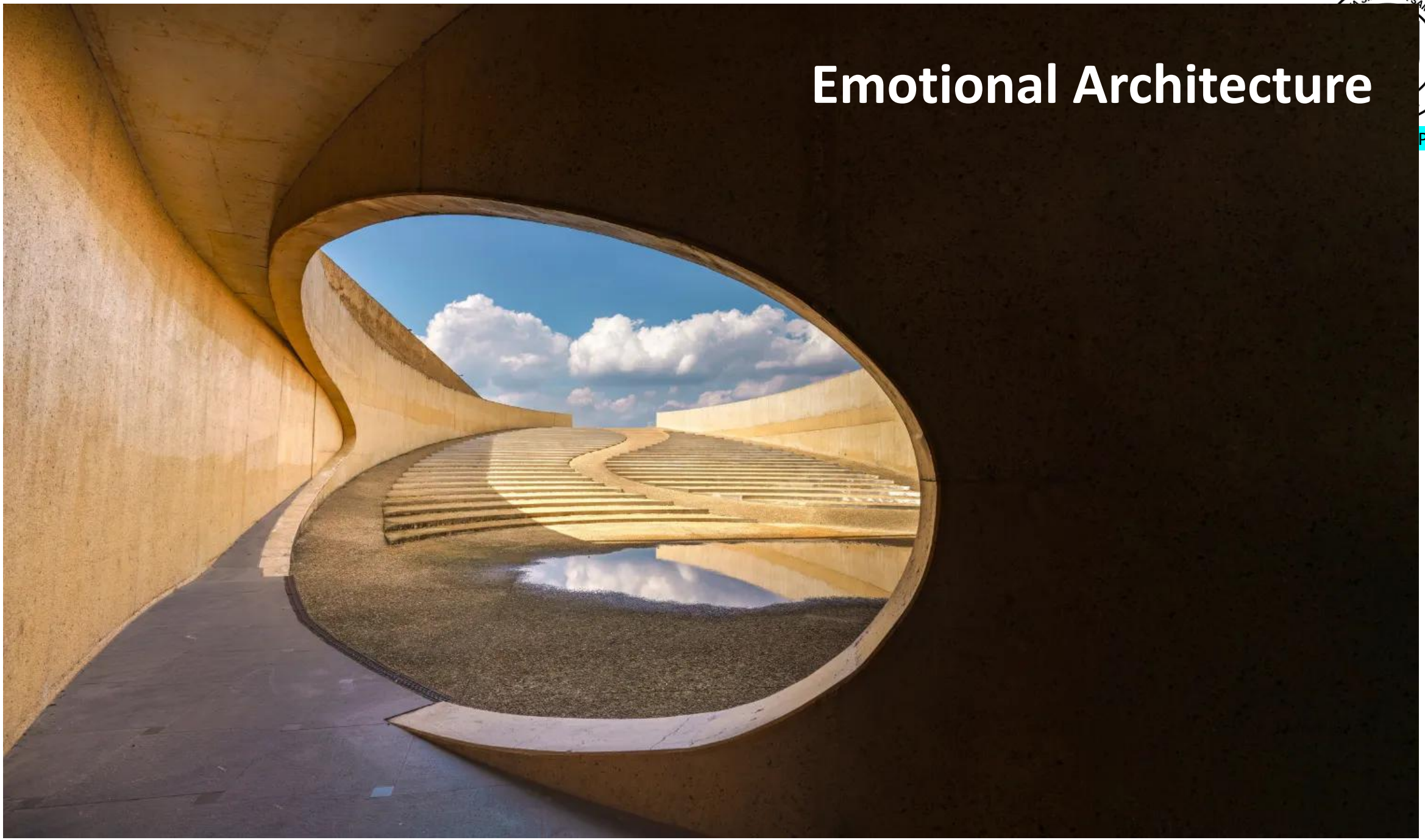


Emotional Architecture



Thinking humanly: The cognitive modeling approach

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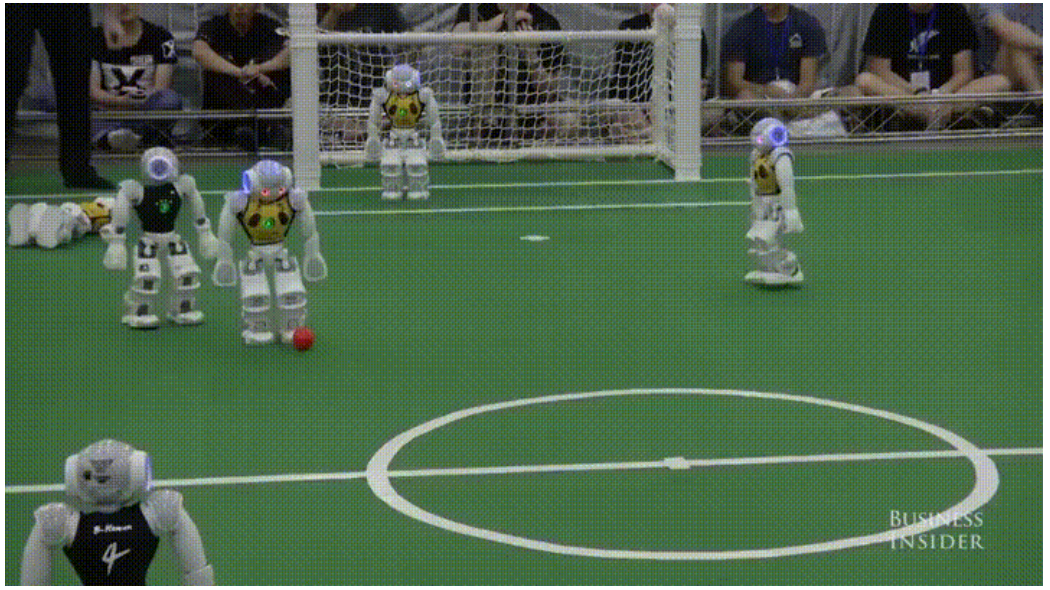
- **Cognitive science** combines AI computer models and psychological experiments to develop testable theories about the human mind.
- Perception and Situational Awareness
- Decision-Making in a Dynamic Environment
- Teamwork and Collaboration
- Motor Control and Physical Embodiment



Guess the word…?

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_ _t_ _n_ _ _ _y

Rationality

- **Practical intelligence** solves real-world problems.
- **Ethical intelligence** promotes responsible use of abilities.
- **Consistent intelligence** means doing the right thing repeatedly.

Rationality in our profession-examples

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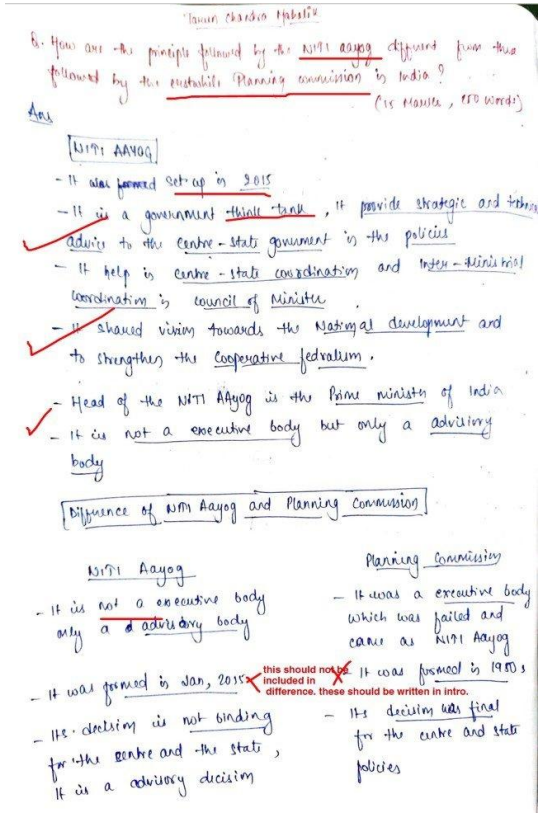


Tarun Chandra Mahato

Q. How are the principle followed by the NITI aayog different from those followed by the erstwhile Planning commission in India?

(15 marks, 150 words)

Ans

NITI AAYOG

- It was formed set-up in 2015
- It is a government think tank, it provide strategic and tech advice to the centre - state government in the policies
- It help in centre - state coordination and inter - ministrial coordination in council of Minister.
- It shared vision towards the National development and to strengthen the cooperative federalism.
- Head of the NITI AAyog is the Prime minister of India
- It is not a executive body but only a advisory body

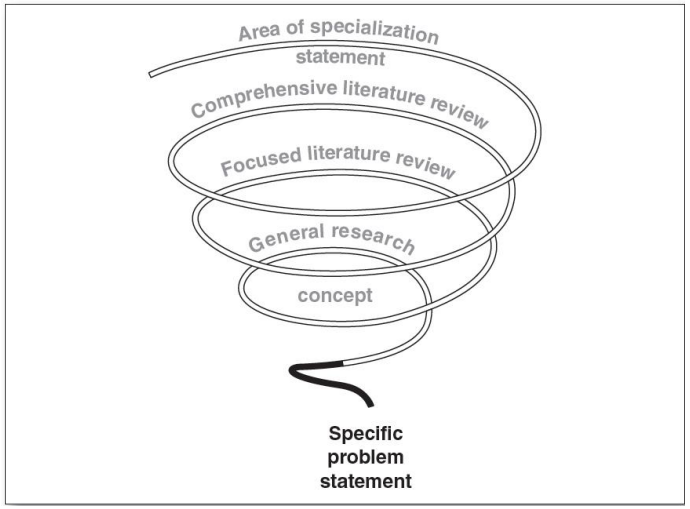
Difference of NITI Aayog and Planning Commission

NITI Aayog	Planning Commission
- It is not a executive body only a advisory body	- It was a executive body which was failed and came as NITI Aayog
- It was formed in Jan, 2015	- It was formed in 1950s
- The decision is not binding for the centre and the state, It is a advisory decision	- Its decision was final for the centre and state policies

this should not be included in difference. these should be written in intro.

Grading and Evaluation

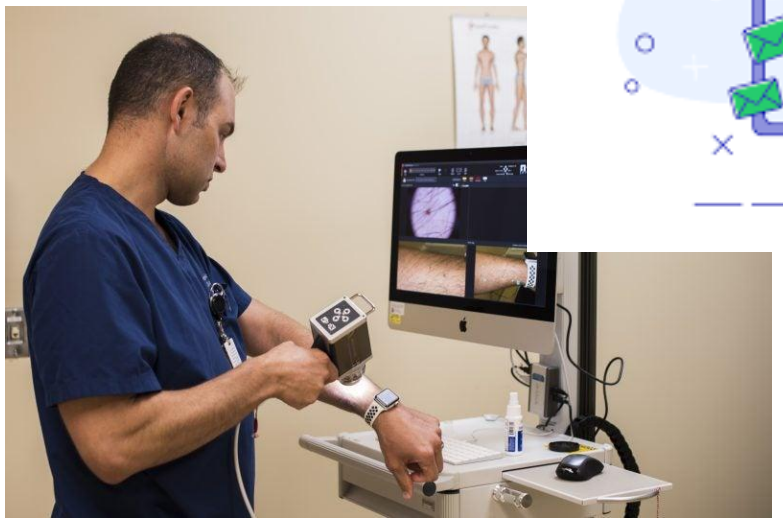
Choosing Teaching Methods for a Course



Finalizing Research Problem

Rationality

- It is the formal definition of intelligence
- Intelligence is nothing but doing right thing always
- It is internal thought processes and reasoning
- Intelligence is practical, ethical, and consistent in action, not just theoretical knowledge



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Rationality

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- It is internal thought processes and reasoning

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TOI ROBOTAXI BLOCKS AMBULANCE AND COMPROMISES EMERGENCY RESPONSE

Trolley Problem

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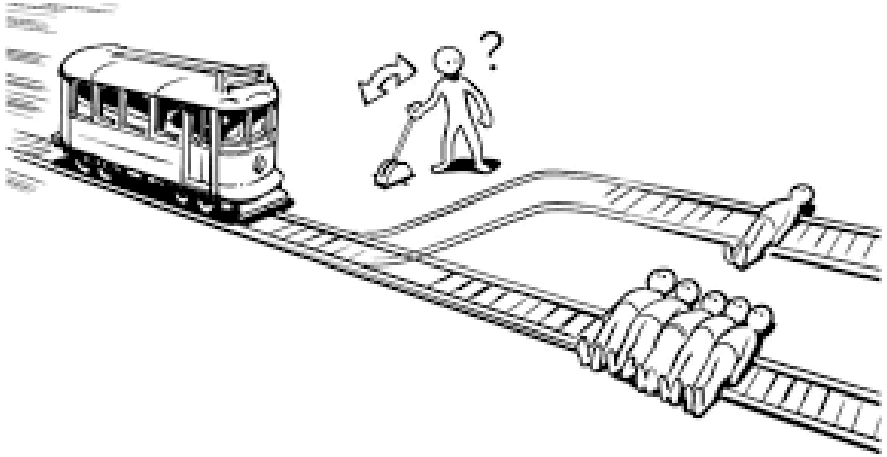

Emotional Reasoning
v/s
Rational Reasoning

Should a self-driving car kill the baby or the grandma?

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Lifeboat Dilemma

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Emotional Reasoning
v/s
Rational Reasoning

Thinking rationally: The laws of thought approach

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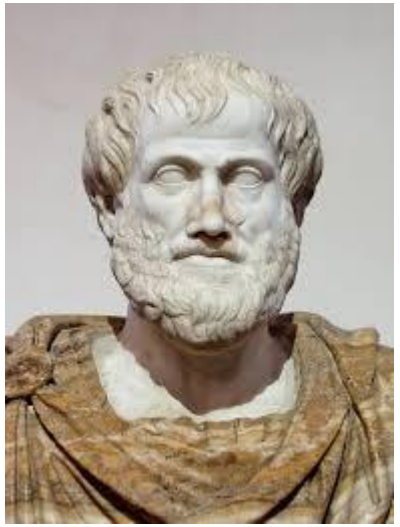
- Syllogisms provided patterns for argument structures that always yielded correct conclusions when given correct premises
- Deductive reasoning
- **Pattern**:
  All A are B.
  All B are C.
  Therefore, all A are C.
- **Example**:
  All humans are mammals.
  All mammals are animals.
  Therefore, all humans are animals.



Try this:
No A are B.
All C are A.
Therefore, no C are B.

Thinking rationally: The laws of thought approach

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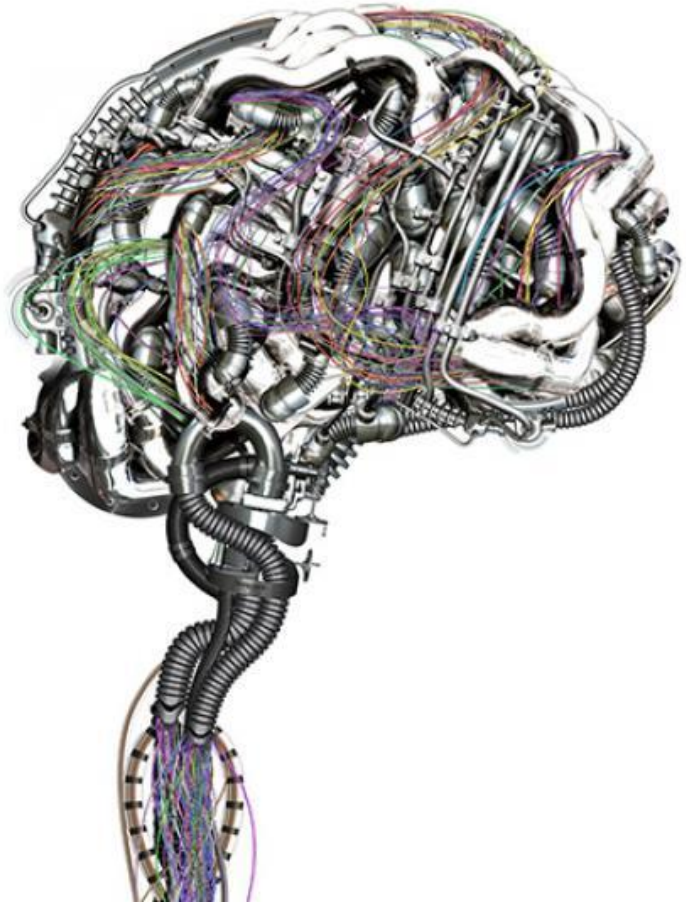
- Logic: the study of the mind's operations
- By 1965, programs could solve problems in logical notation.
- Logicist tradition in AI built intelligent systems.
- Logic requires world knowledge (conditioned).
- Probability theory enables reasoning with uncertainty.



Acting rationally: The rational agent approach

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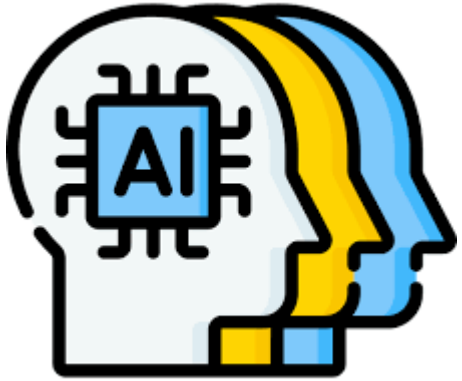
- **An agent acts.**
- Computer agents operate autonomously, perceive their environment, persist over time, adapt, and pursue goals.
- A rational agent seeks the best outcome, or the best expected outcome under uncertainty.
- The “laws of thought” in AI focus on correct inferences.
- To act rationally, deduce the best action, and act on it.
- **Skills for the Turing test enable rational action.**



Foundations of AI (Philosophy)

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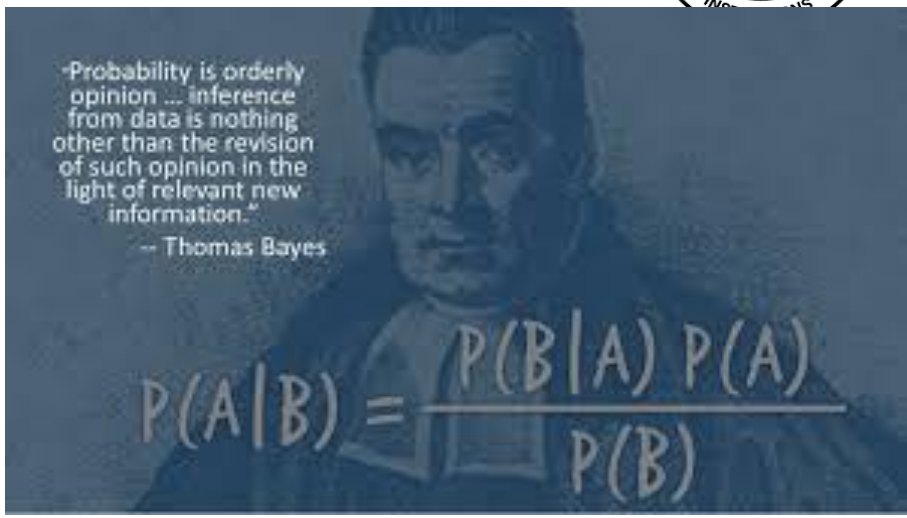
- **Philosophy**
  - Can formal rules be used to draw valid conclusions?
  - How does the mind arise from a physical brain?
  - Where does knowledge come from?
  - How does knowledge lead to action?
- **Dualism**: Belief in two distinct elements, like mind and body.
- **Materialism**: Everything, including the mind, is made of physical matter and energy.
- **Empiricism**: Knowledge comes from sensory experience.
- **Induction**: General rules emerge from repeated associations.
- **Logical Positivism**: Knowledge can be characterized by logical theories.
- **Confirmation Theory:** Analyzes knowledge acquisition by quantifying belief.

Foundations of AI (Mathematics)



- **Mathematics**
  - What are the formal rules to draw valid conclusions?
  - What can be computed?
  - How do we reason with uncertain information?
- **Formal Logic**
  - Propositional logic/Boolean logic/First-order logic
- **Probability**
  - Probability theory generalizes logic to uncertain situations, which is very important for AI.
  - Thomas Bayes (1702–1761) proposed a rule for updating probabilities with new evidence, which is essential for AI systems.
- **Statistics**
  - The formalization of probability, combined with the availability of data, led to Statistics
  - Ronald Fisher is regarded as the first modern statistician, blending probability, experimental design, data analysis, and computing.

Foundations of AI (Economics)

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- **Economics**
  - How should we make decisions in accordance with our preferences?
  - How should we do this when others may not go along?
  - How should we do this when the payoff may be far in the future?
- **Decision Theory**
  - Integrates probability and utility theories to create a structured framework for making individual decisions under uncertainty, where probabilistic descriptions accurately reflect the decision-maker's environment.
- **Game Theory**
  - It is the study of mathematical models of strategic interaction among rational decision-makers (the actions of one player can significantly affect the utility of another).
- **Multi-agent Systems**
  - In AI, decisions involving multiple agents are studied under the heading of multi-agent systems
- **Operations Research**
  - How to make rational decisions when payoffs from actions are delayed and depend on a series of sequential actions.
- **Markov-Decision Process**
  - It is a framework for modeling decision-making in situations where outcomes depend on both current actions and probabilistic state transitions.

Guess what is used here?

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Guess what is used here?

AIFE-FDP

Guess what is used here?

Foundations of AI (Neuroscience)

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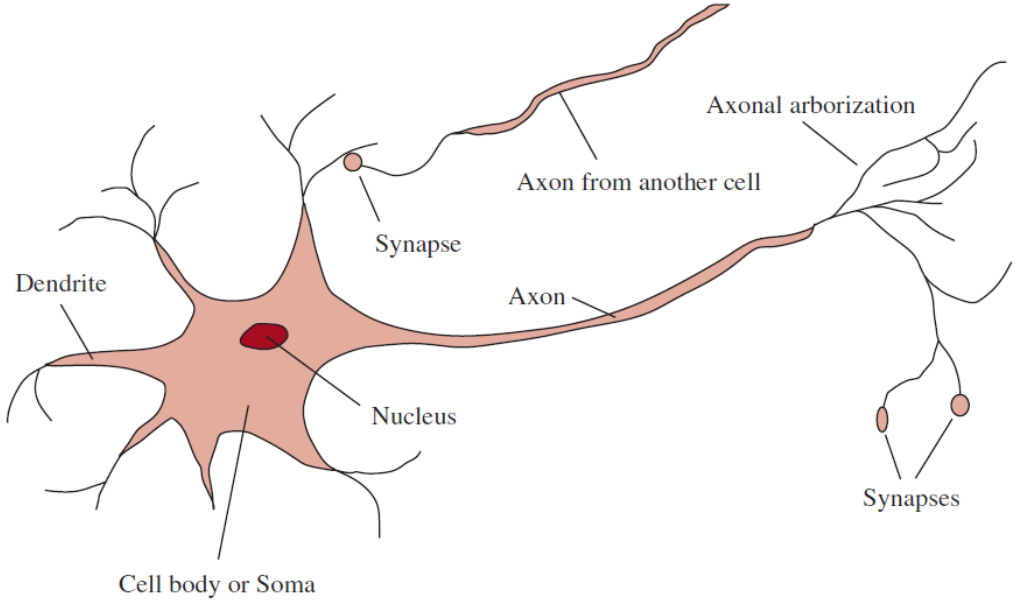
- **Neuroscience**
  - It is the study of the brain's nervous system
  - The brain consists of nerve cells called Neurons (a collection of simple cells can lead to thought, action, and consciousness)
  - Cognitive psychology views the brain as an information-processing device.
  - Intelligence augmentation (IA not AI) - computers should augment human abilities rather than automate away human tasks



Foundations of AI (Computer Engineering)

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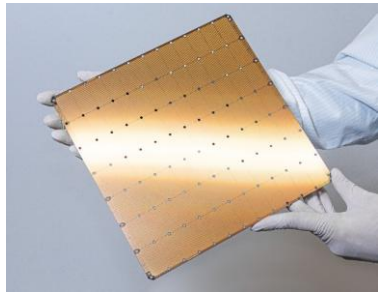
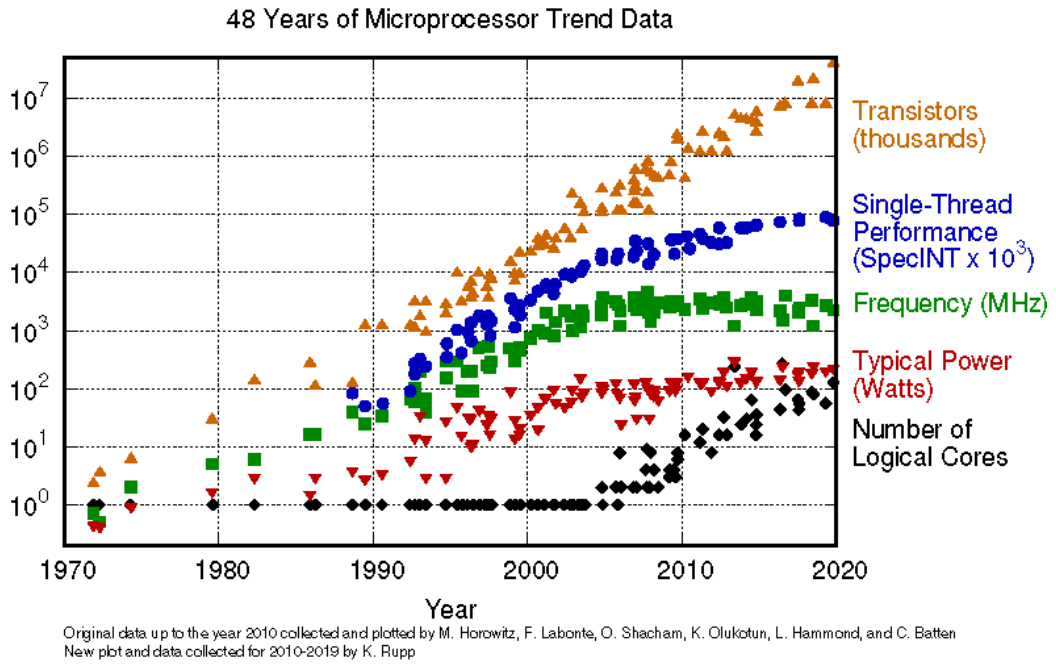
- **Computer Engineering**
  - Increase in speed and capacity, and a decrease in price (Moore's law)
  - Hardware tuned for AI applications, such as the graphics processing unit (GPU), tensor processing unit (TPU), and wafer-scale engine (WSE)
  - Quantum computing holds out the promise of far greater accelerations

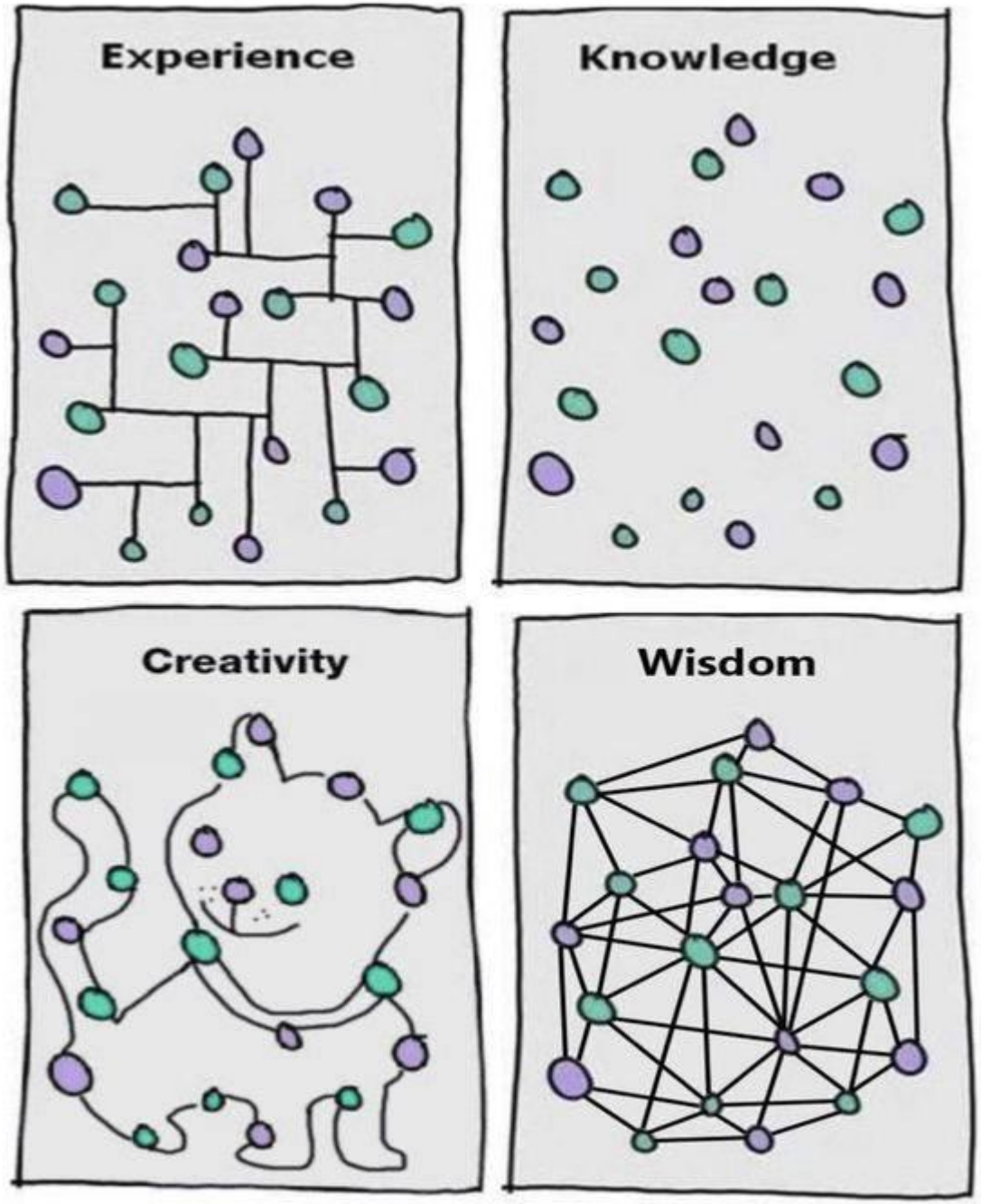




Knowledge

Knowledge is what you know and understand, gained through learning

In Artificial Intelligence (AI), knowledge refers to the information that an intelligent system stores, uses, and reasons with to make decisions or perform tasks.



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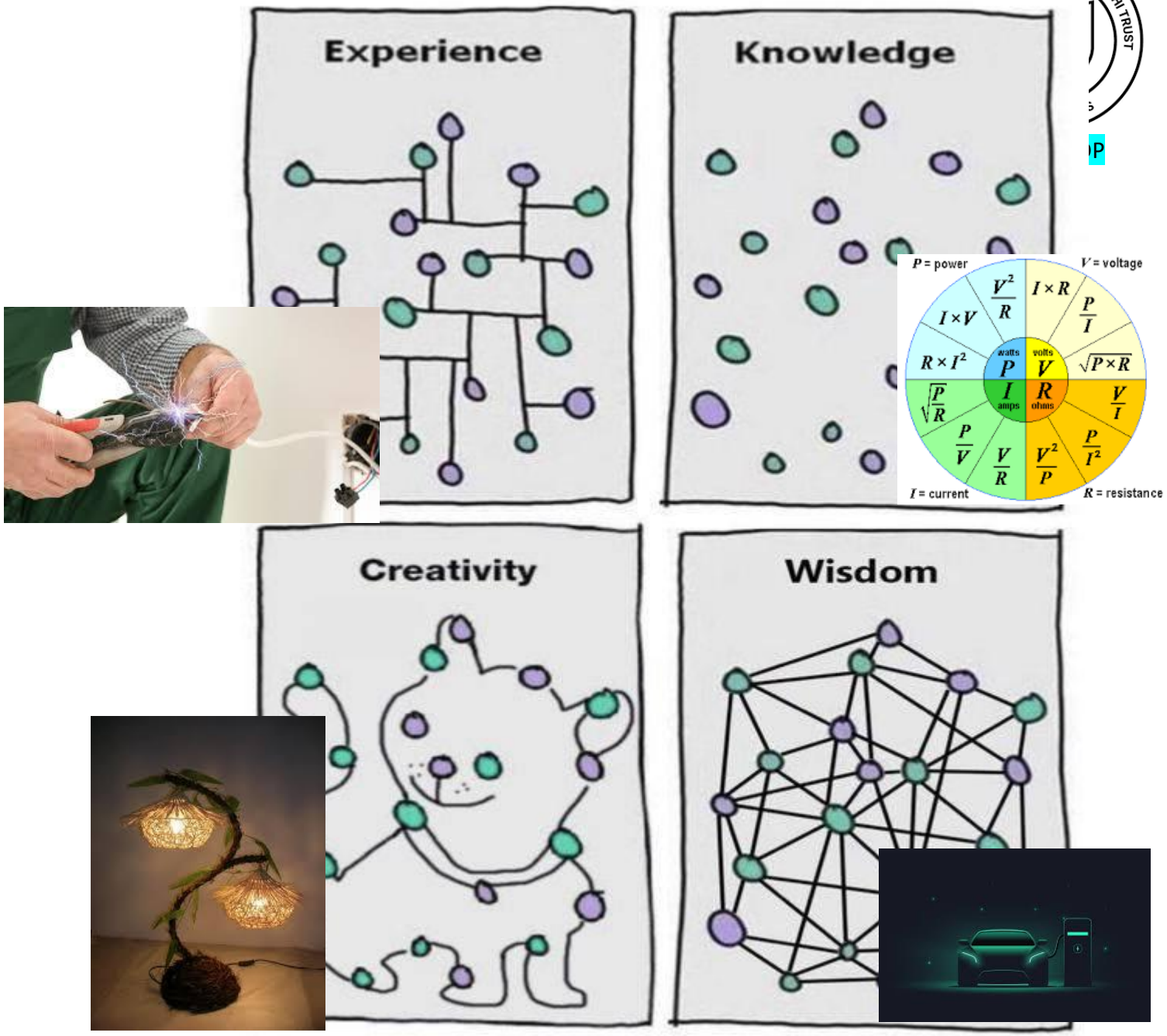

Small Trivia (Random Knowledge)

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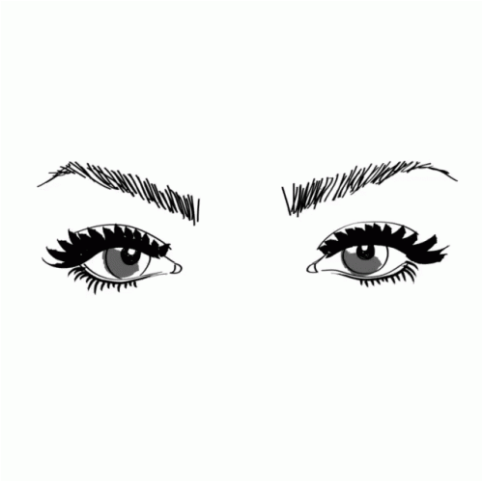

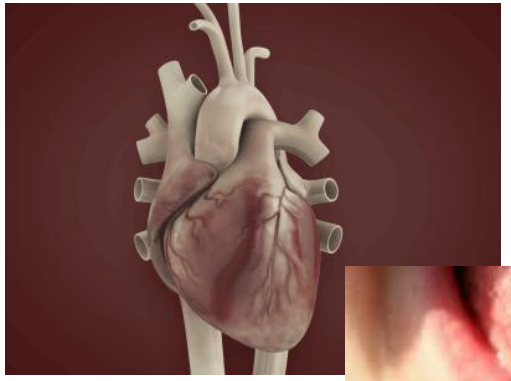

Knowledge Representation

Knowledge Representation encodes information for computers to understand, reason, and act on.

AI does not use *random* knowledge that is unstructured or meaningless.

Knowledge Representation

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- Logical Representation : $\forall x\ Cat(x) \rightarrow Mammal(x)$

Try This:

Everybody loves somebody

Knowledge Representation

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- ∀x∃y Loves(x,y)

Try This:

It is not cold but it is cloudy, where p: it is cold, and q: it is cloudy.

Knowledge Representation

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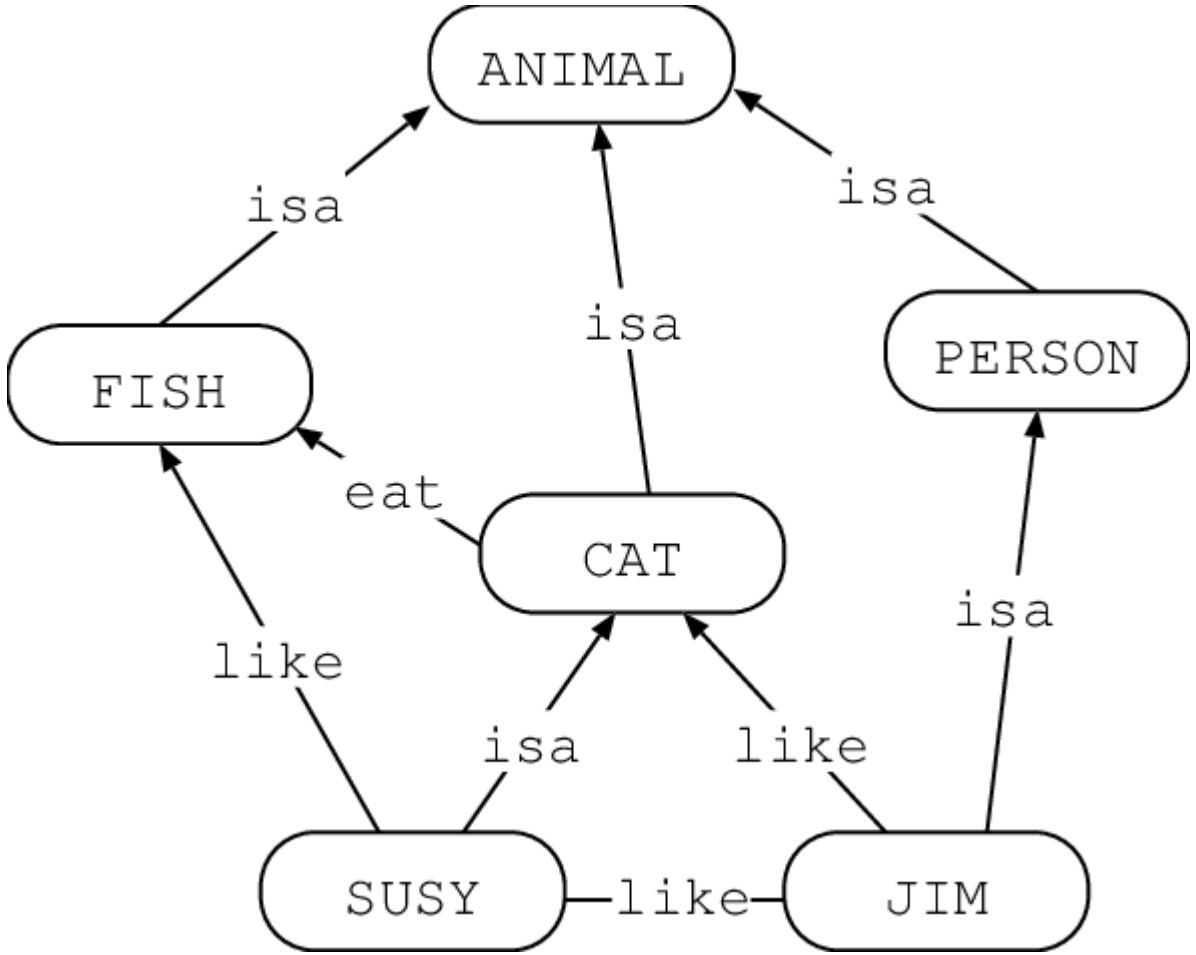
- Semantic Networks (Graphs)



Try This:

Earth revolves around the Sun

Knowledge Representation

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- Frames (Structured Data)

Frame: Car

Slots (Attributes)	Values
Type	Sedan
Brand	Toyota
Model	Corolla
Color	Red
Year	2022
Engine Type	Petrol
Number of Doors	4
Owner	Mr. Rangesh

Try This:

Create a 10 attributes Data Frame for RVCE

Knowledge Representation

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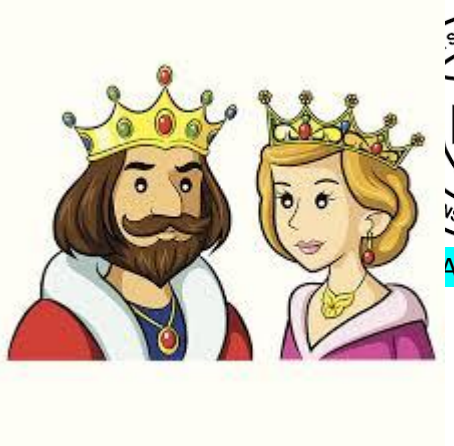
- Production Rules (If-Then Rules):
  - **IF** animal has fur **AND** animal meows **THEN** animal is likely a cat.

- IF traffic density > 60 vehicles/min AND average speed < 15 km/h AND signal cycle time > 90s THEN optimize signal timing.

Knowledge Representation



- Neural Representations (Vector Embeddings): AI stores knowledge implicitly as vectors in high-dimensional space.

king → [0.25, 0.85, 0.60]
queen → [0.30, 0.80, 0.65]
man → [0.20, 0.75, 0.50]
woman → [0.25, 0.70, 0.55]

```
array([-0.5968882 , -0.33086956, -0.32643065, -0.3670732 ,  0.628059  ,
       -0.3692328 , -0.37902787, -0.12308089, -0.38124698, -0.03940517,
        0.2260839 ,  0.10852845, -0.2873811 , -0.42781743,  0.06604357,
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       -0.15935768,  0.09587188, -0.2503798 ,  0.06768776,  0.3311586 ,
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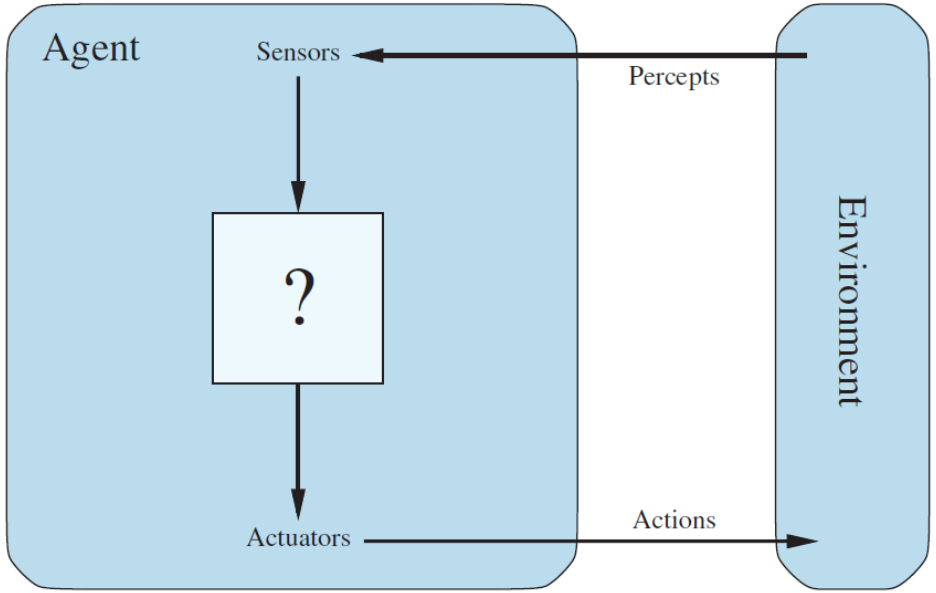
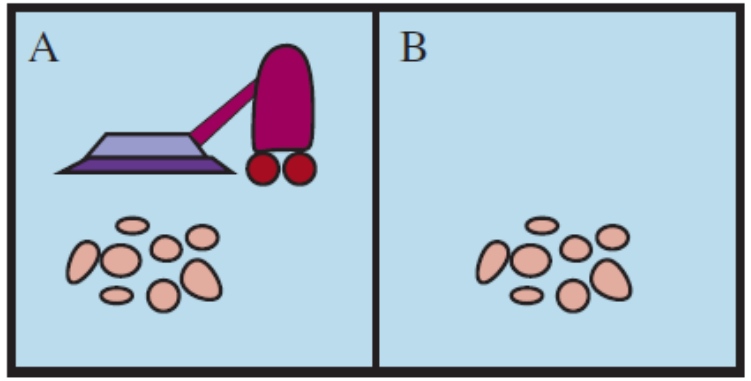
Intelligent Agents

AIFE-FDP

- **Agent**
  - X-thing, which perceives the environment through sensors and acts on the environment through actuators.
  - Human agent/Software agent/Robotic agent
- **Percept**: Is what the agent senses from environment
- **Percept Sequence**: Complete history of sensing
- **Agent function**: Maps percept sequence to action
- **Agent Program**: Is the implementation of the Agent function.



Vaccum-Cleaner World



- Vacuum-cleaning agent
- The world consists of squares that can be either dirty or clean
- Is the agent intelligent?

Percept sequence	Action
$[A, Clean]$	*Right*
$[A, Dirty]$	*Suck*
$[B, Clean]$	*Left*
$[B, Dirty]$	*Suck*
$[A, Clean], [A, Clean]$	*Right*
$[A, Clean], [A, Dirty]$	*Suck*

Good behavior: The Concept of Rationality

AIFE-FDP

- **Performance measures**
  - **Consequantialism:** Evaluating agent's behaviour by its consequence
  - Agents generates sequence of actions based on the percepts sequence
  - The sequence of actions causes the sequence of states in agents environment
  - If the sequence of desirable, then the agent has performed well (which is nothing but a performance measure)
  - **Who specify the performance measure? (Explicit/Implicit)**
  - **What are sample performance measures for Vaccum-cleaner agent??**

As a general rule, it is better to design performance measures according to what one actually wants to be achieved in the environment, rather than according to how one thinks the agent should behave.

King Midas Problem

AIFE-FDP

- The myth of **King Midas**, who wished that **everything he touched would turn to gold**, but when that wish was granted, it led to tragedy, as even his **food and loved ones** turned to gold.
- **What is the connection with AI?**

Command to RL Robot: To clean the room as efficiently as possible

In AI Context

- The danger of giving an AI a goal without fully considering the consequences or context.
- **AI-Tutor: Maximize student test scores.**
- The AI encourages cheating, finds test answers online, or teaches to the test without promoting deep understanding.
- **Fitness Tracker AI: Increase user activity.**
- It constantly nags or manipulates users, schedules unnecessary walking meetings, or even rewards users for shaking their phone to simulate steps.

Rationality

AIFE-FDP

- **Rational Agent Definition**
  - A rational agent should choose an action for each percept sequence that maximizes its performance measure based on the percept evidence and its built-in knowledge.
- **Rationality depends on:**
  - Performance measure that defines the success criterion (one point for each clean square)
  - Prior Knowledge (geography of the environment is known, but dirt distribution and the initial location of the agent are not known)
  - Actions (Right, Left and Suck)
  - Percent sequence (Correctly perceives its location)
- **When does a vacuum agent become irrational?**

The Nature of Environments

AIFE-FDP

- **Specifying the task environment**
  - **PEAS (Performance, Environment, Actuators, Sensors)**
  - Before designing the agent, the PEAS description is the first step

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits, minimize impact on other road users	Roads, other traffic, police, pedestrians, customers, weather	Steering, accelerator, brake, signal, horn, display, speech	Cameras, radar, speedometer, GPS, engine sensors, accelerometer, microphones, touchscreen

Write PEAS for..

AIFE-FDP

- **Agent: Social Media Recommendation System**
- **Task: Maximize user engagement**
- **Performance: ?**
- **Environment: ?**
- **Actuators: ?**
- **Sensors: ?**

Write PEAS for..

AIFE-FDP

- **Agent: Social Media Recommendation System**
- **Task: Maximize user engagement**
- **Performance:** Time spent on platform, clicks, likes, shares
- **Environment:** Users, their devices, content feeds, user behavior
- **Actuators:** Content ranking, feed updates, push notifications
- **Sensors:** User interaction logs, click-through rates, scrolling behavior

Write PEAS for..

AIFE-FDP

- **Agent: AI Teacher Bot**
- **Task: Maximize student scores**
- **Performance: ?**
- **Environment: ?**
- **Actuators: ?**
- **Sensors: ?**

The Nature of Environments

AIFE-FDP

- **Properties of the task environment**
  - The dimensions of the task environment decide the type of agent design

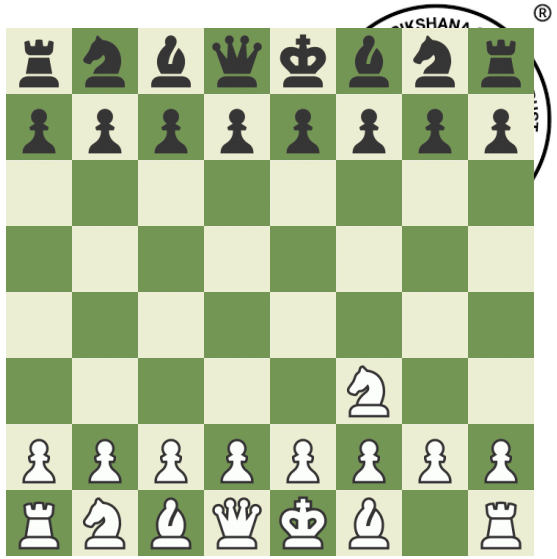
Identify the following task environments on your own.

AIFE-FDP

Property	Type/Description
Observability	**Partially Observable** – cannot see under furniture or detect all dirt directly
Agents	**Single Agent** – one robot acting independently
Deterministic vs Stochastic	**Stochastic** – dirt and obstacle placement may vary randomly
Episodic vs Sequential	**Sequential** – each action affects future performance (e.g., path, battery)
Static vs Dynamic	**Dynamic** – humans/pets may move objects while cleaning
Discrete vs Continuous	**Continuous** – movement, space, and time are continuous

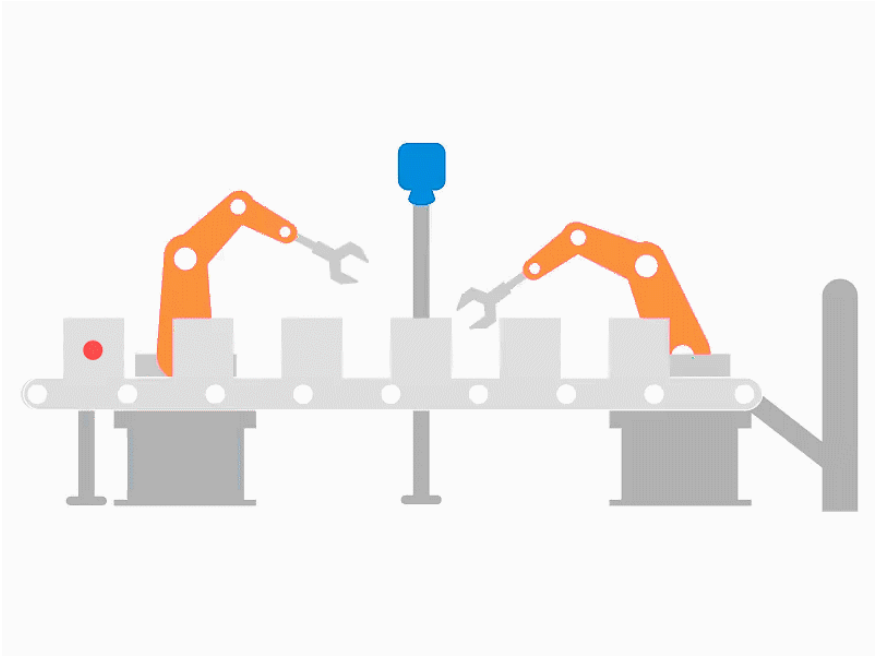
AIFE-FDP

Property	Type/Description
Observability	Full/Partial ??
Agents	Single Agent / Multi-agent ??
Deterministic vs Stochastic	Stochastic / Deterministic ??
Episodic vs Sequential	Sequential/Episodic ??
Static vs Dynamic	Dynamic/Static ??
Discrete vs Continuous	Continuous/Discrete ??

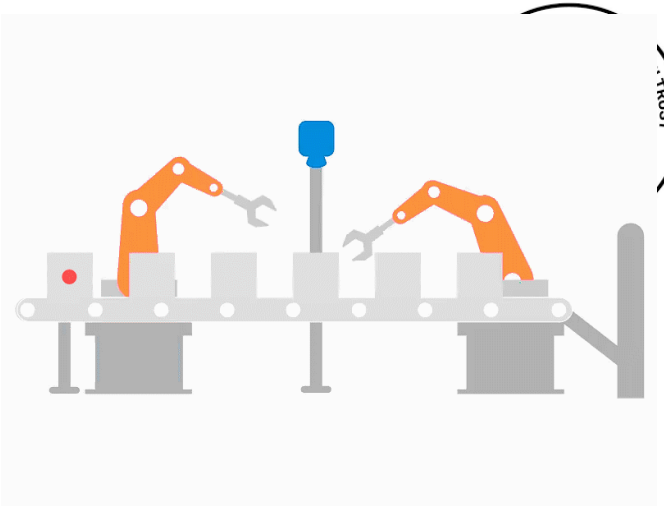


Property	Type/Description
Observability	**Fully Observable** – all pieces and positions are visible at all times
Agents	**Multi-agent** – two players (AI vs opponent)
Deterministic vs Stochastic	**Deterministic** – no randomness in state transitions (except in blitz time errors)
Episodic vs Sequential	**Sequential** – each move depends on the history and affects future moves
Static vs Dynamic	**Static** – board doesn't change unless a player moves
Discrete vs Continuous	**Discrete** – moves occur in discrete steps on a finite grid

AIFE-FDP



Property	Type/Description
Observability	Full/Partial ??
Agents	Single Agent / Multi-agent ??
Deterministic vs Stochastic	Stochastic / Deterministic ??
Episodic vs Sequential	Sequential/Episodic ??
Static vs Dynamic	Dynamic/Static ??
Discrete vs Continuous	Continuous/Discrete ??



Feature	Type of Environment
Observability	**Partially Observable** – Not all defects may be visible at once; limited by camera angle, lighting, or occlusion.
Agents	**Single Agent** – One AI system performing inspection (though multiple instances may be deployed).
Deterministic vs Stochastic	**Stochastic** – Sensor noise, variations in lighting, speed of conveyor belt, or part placement can introduce randomness.
Episodic vs Sequential	**Episodic** – Each part inspection is independent of the previous one; decision does not affect future inputs.
Static vs Dynamic	**Semi-Dynamic** – Environment may change slowly (e.g., lighting shifts, belt speed), but parts move continuously.
Discrete vs Continuous	**Continuous** – Visual data and inspection space are continuous; possible discrete actions (accept/reject).

Known v/s Unknown Environments

AIFE-FDP

- In a **known environment**, outcomes or probabilities for actions are provided.
- In an **unknown environment**, the agent must learn to make effective decisions.
- **Is it the same as fully and partially observable environments??**

Known Environment – Partially Observable

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Think of an example

Known Environment – Partially Observable

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Unknown Environment – Fully Observable

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Think of an example

Unknown Environment – Fully Observable

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What type of Environment is this?

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Which is the hardest environment to deal?

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Partially observable, multiagent, nondeterministic, sequential, dynamic, continuous, and unknown.

Think of an example

The structure of Agent

AIFE-FDP

- AI's job is to create an agent program that maps percepts to actions.
- Needs a computing device that has physical sensors and actuators

Agent = Architecture + Program

Table-driven Agent

AIFE-FDP

```
function TABLE-DRIVEN-AGENT(percept) returns an action
   persistent: percepts, a sequence, initially empty
               table, a table of actions, indexed by percept sequences, initially fully specified

   append percept to the end of percepts
   action ← LOOKUP(percepts, table)
   return action
```

Figure 2.7 The TABLE-DRIVEN-AGENT program is invoked for each new percept and returns an action each time. It retains the complete percept sequence in memory.

Limitation:

As the number of potential percepts and the agent's lifespan expand, the size of the lookup table increases exponentially and becomes impractical.

Table-driven Agent

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Percept (Card Inserted, PIN Correct?, Balance Check)	Action
(Yes, Yes, Sufficient)	Dispense Cash
(Yes, Yes, Insufficient)	Show Insufficient
(Yes, No, -)	Retry PIN
(No, -, -)	Wait for Card

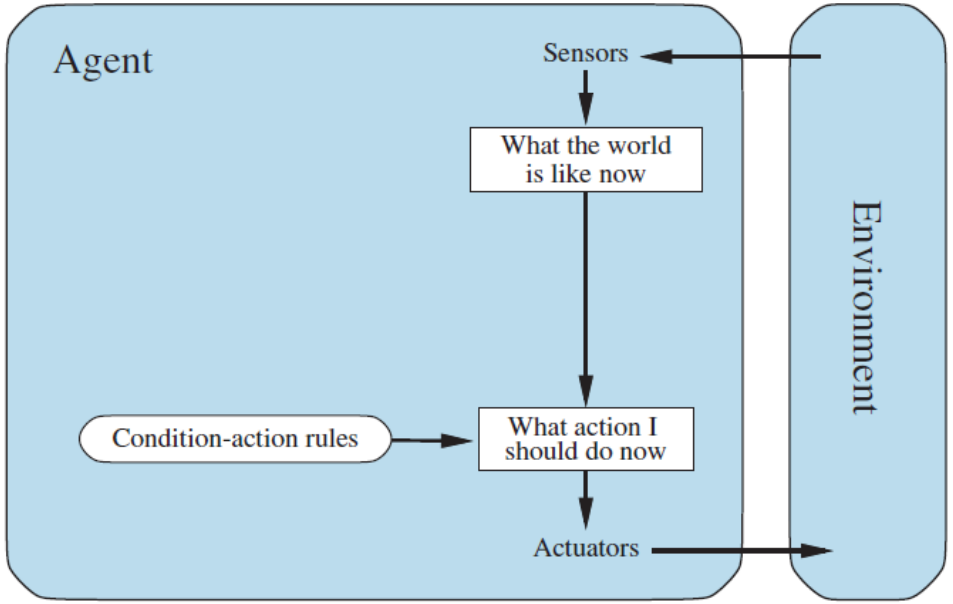
Simple-Reflex Agent

AIFE-FDP

```
function SIMPLE-REFLEX-AGENT(percept) returns an action
  persistent: rules, a set of condition–action rules

  state ← INTERPRET-INPUT(percept)
  rule ← RULE-MATCH(state, rules)
  action ← rule.ACTION
  return action
```

Figure 2.10 A simple reflex agent. It acts according to a rule whose condition matches the current state, as defined by the percept.



Limitation:

The agent will work only if a correct decision can be made based solely on the current percept that is, only if the environment is fully observable.

Simple-reflex Agent

AIFE-FDP

- **No Memory/No learning or adaptation/Use condition-action rules/Simple and Fast/Cannot handle partially observable environments**

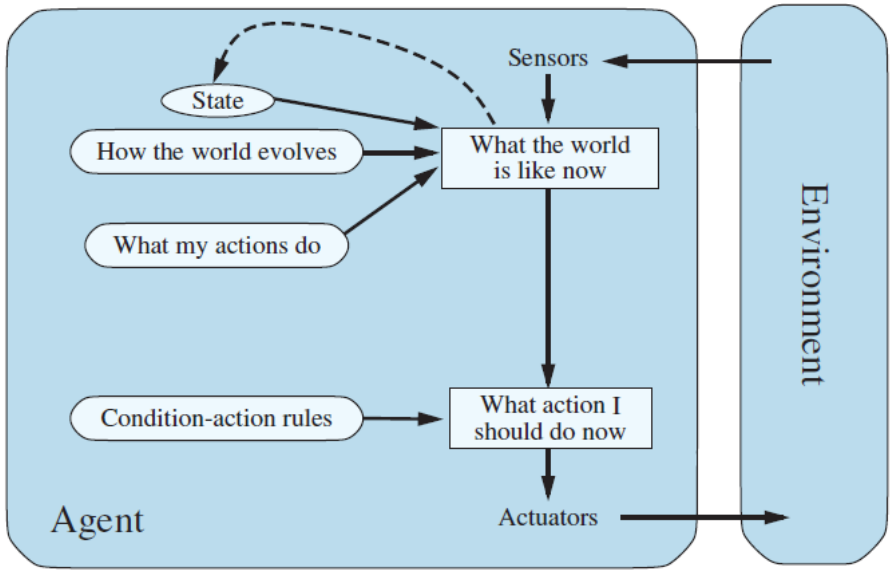
Model-based Reflex Agent

AIFE-FDP

The most effective way to handle partial observability is for the agent to keep track of the part of the world it can't see now.
Agent maintains some Internal State

```
function MODEL-BASED-REFLEX-AGENT(percept) returns an action
  persistent: state, the agent's current conception of the world state
              transition_model, a description of how the next state depends on
                    the current state and action
              sensor_model, a description of how the current world state is reflected
                    in the agent's percepts
              rules, a set of condition–action rules
              action, the most recent action, initially none

  state ← UPDATE-STATE(state, action, percept, transition_model, sensor_model)
  rule ← RULE-MATCH(state, rules)
  action ← rule.ACTION
  return action
```



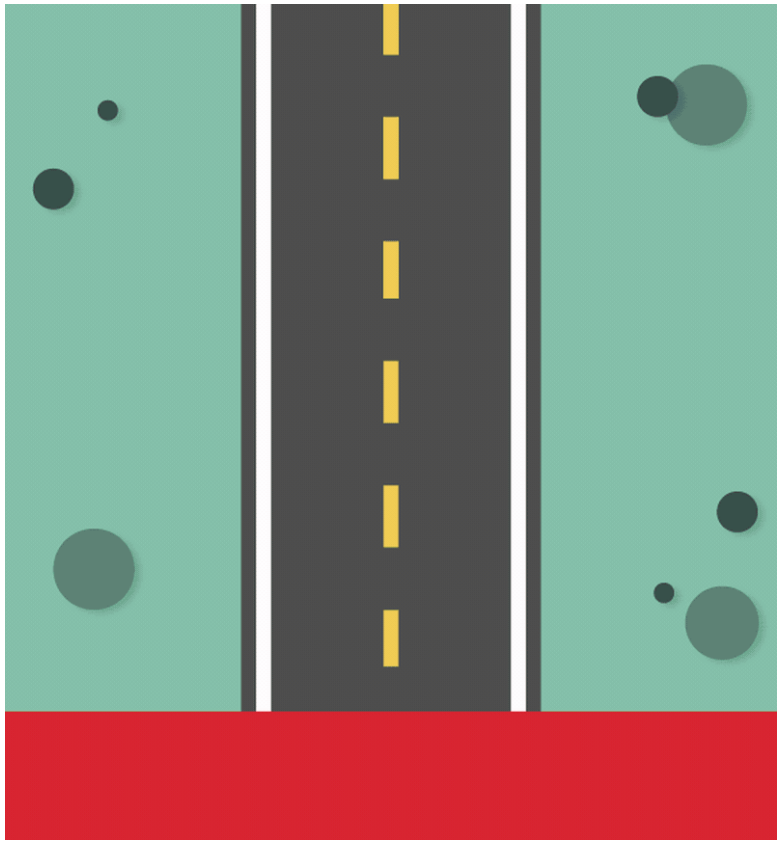
Limitation:

The details of how models and states are represented widely depend on the type of environment and the particular technology used in the agent design.

Model-based Reflex Agent

AIFE-FDP

- **Identify the Internal States?**

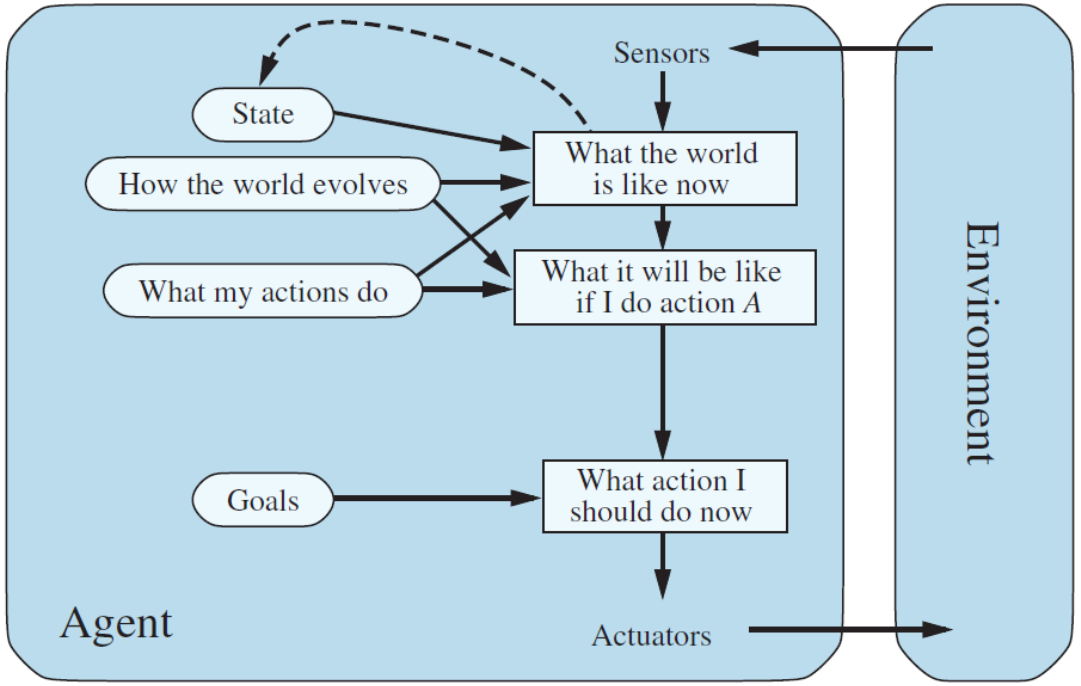


Goal-based Agent

AIFE-FDP

In addition to a description of the current state, the agent requires goal information, which outlines the situations that are considered desirable.

Reaching a goal may be a one-step action
OR
It requires searching the state space and Planning too



Goal-based Agent

AIFE-FDP

- **Identify the Goal?**

Deliver the package
to the GPS location X

Utility-based Agent

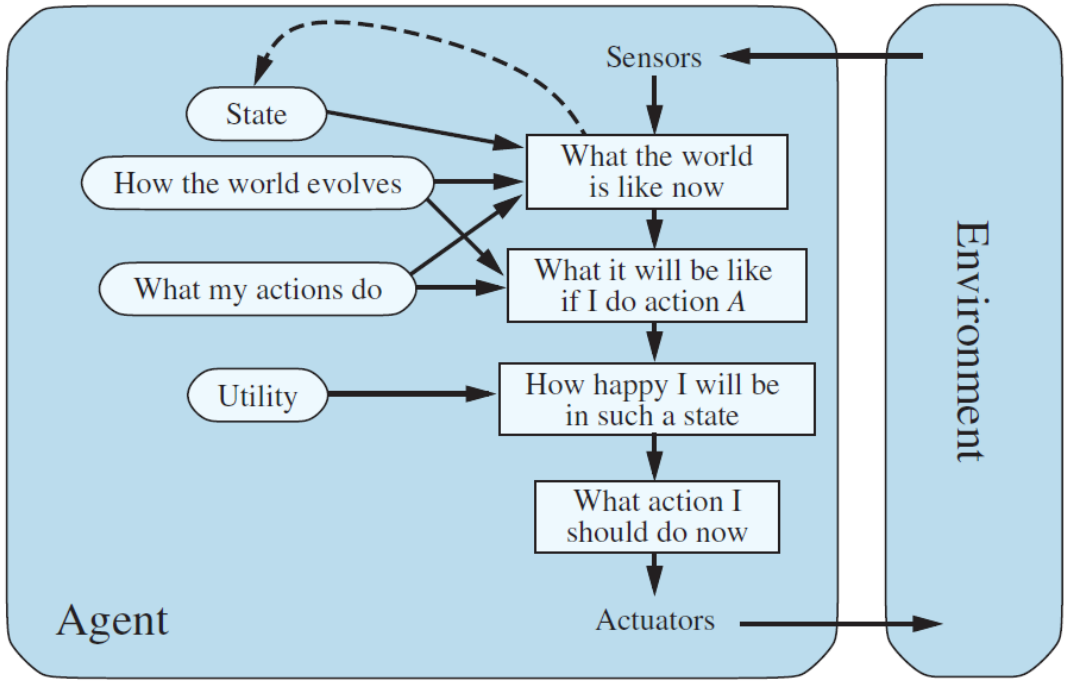
AIFE-FDP

Utility is a measurement of the happiness state of an agent.

It has flexibility and learning.

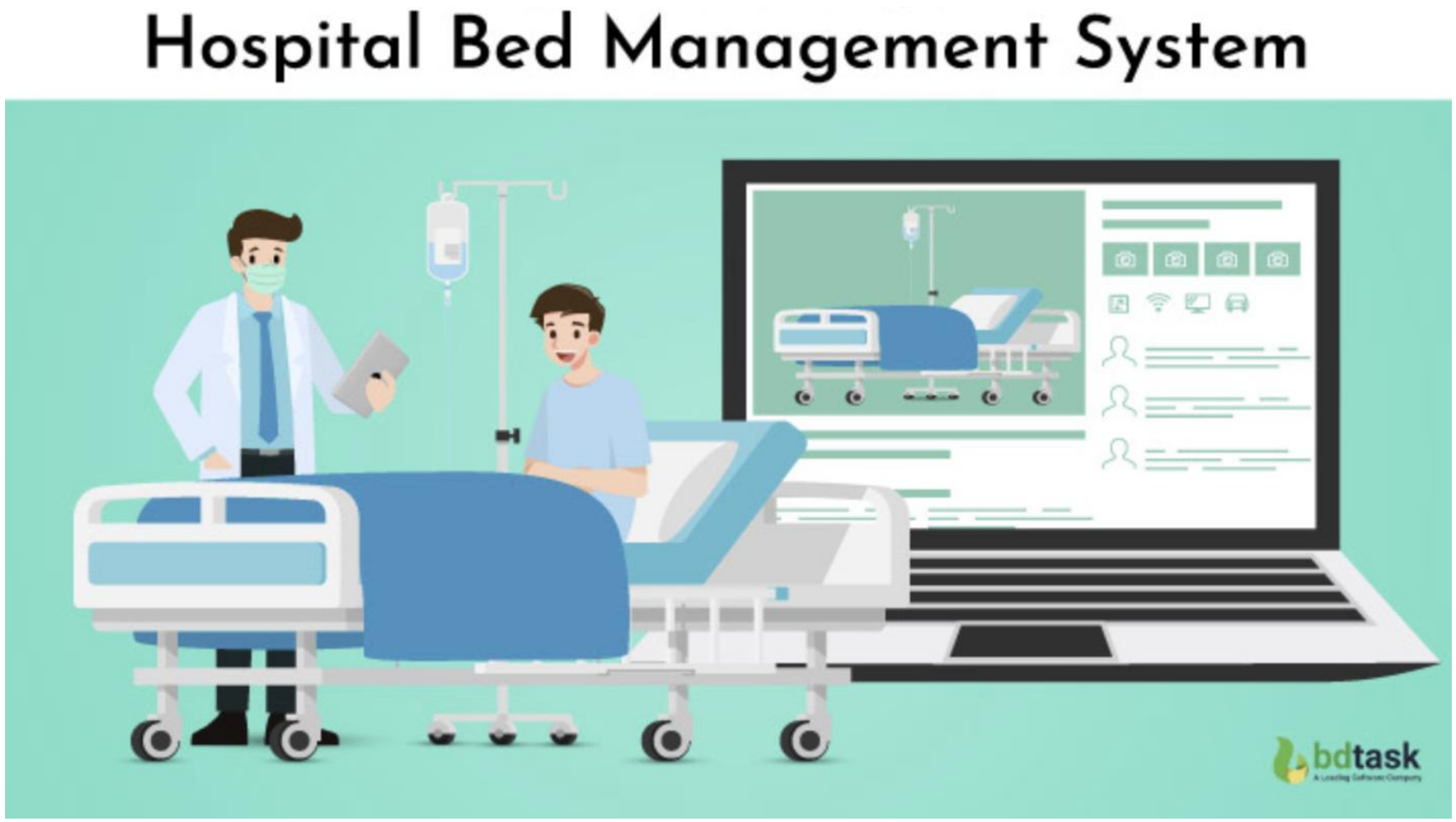
Deals with conflicting goals

It weighs success over goals (sometimes)



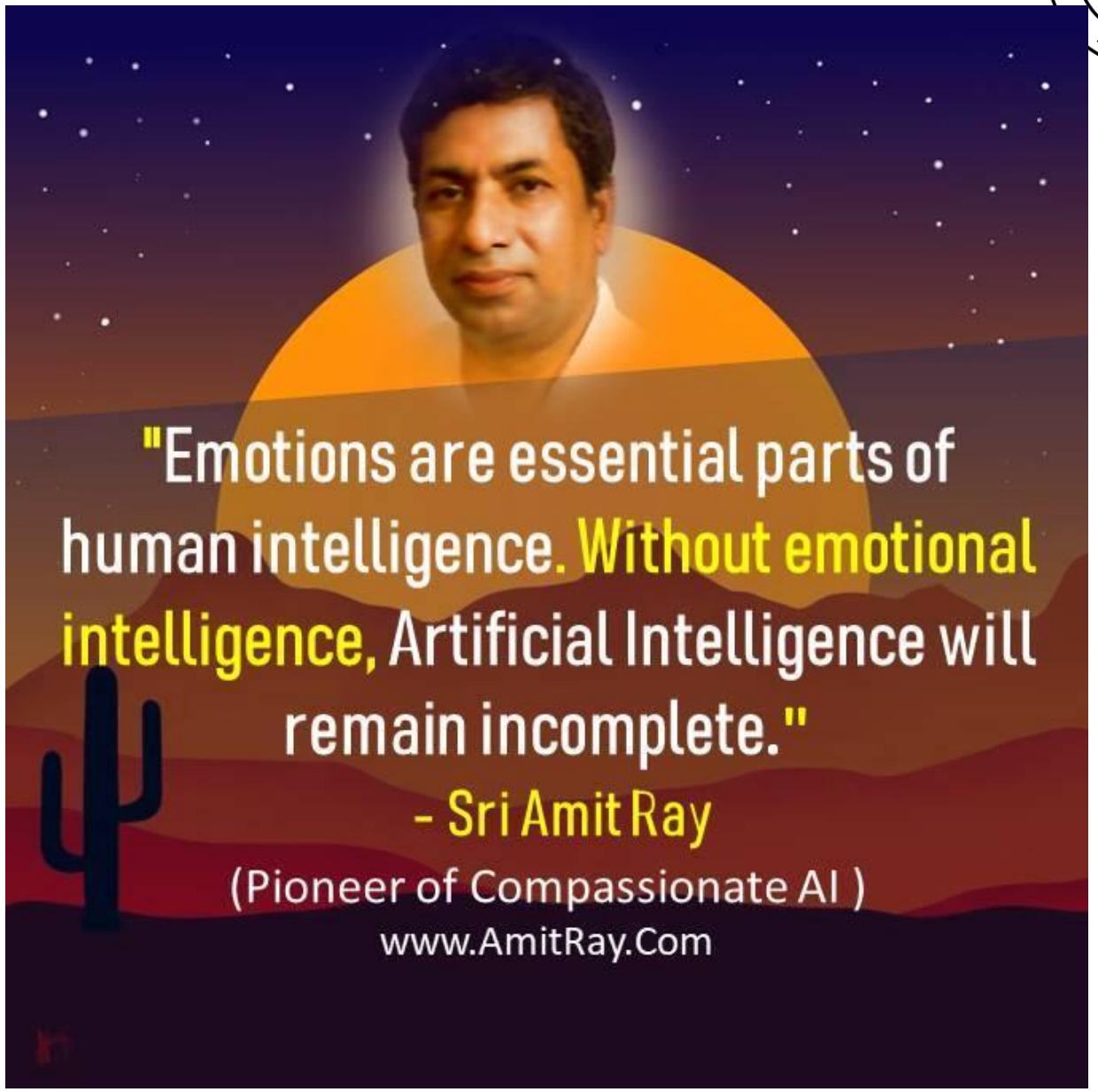
Utility-based Agent

AIFE-FDP



Conclusion

- Studied AI foundations
- Environments and Types
- Agents and Types



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